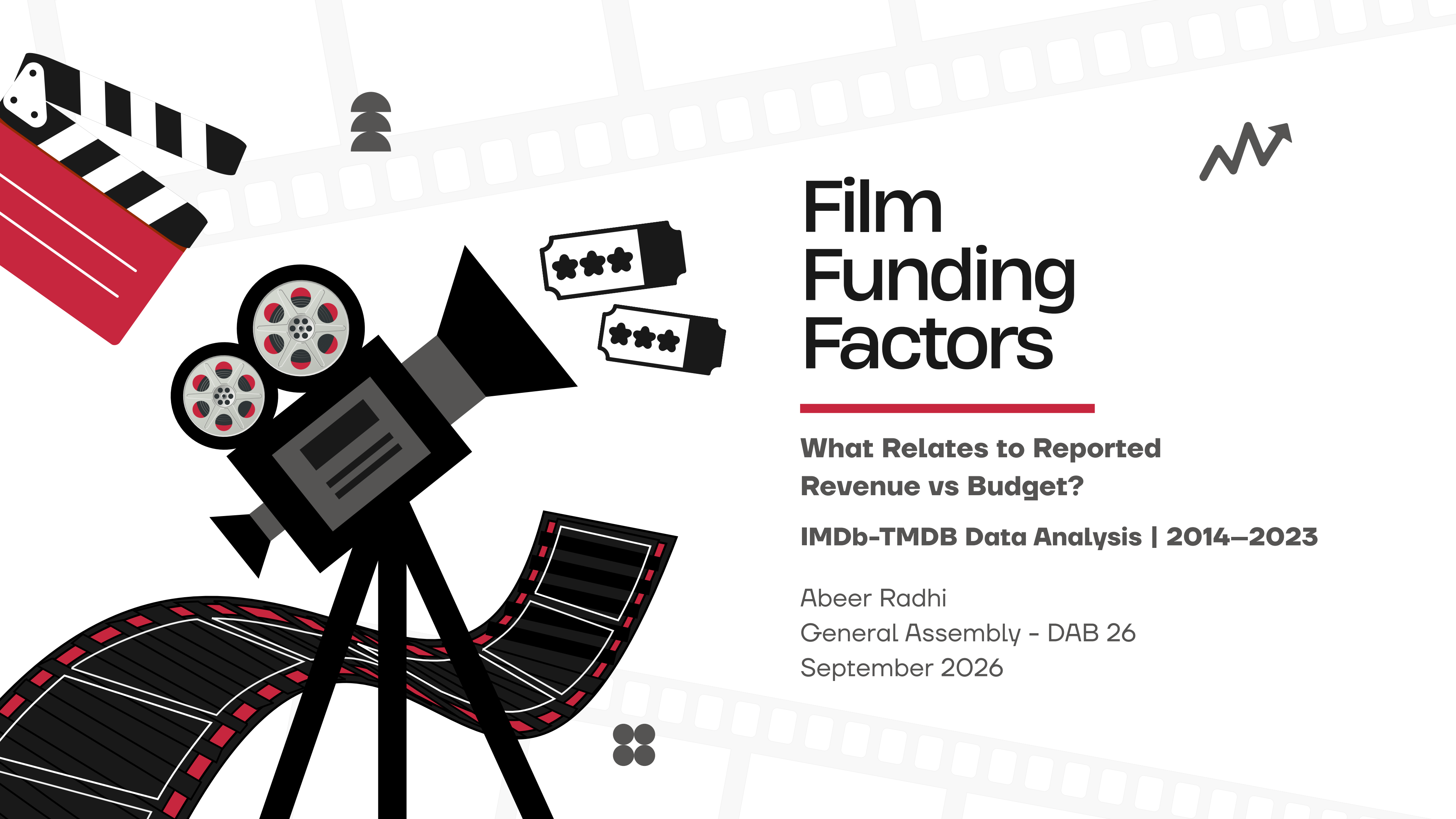




Film Funding Factors

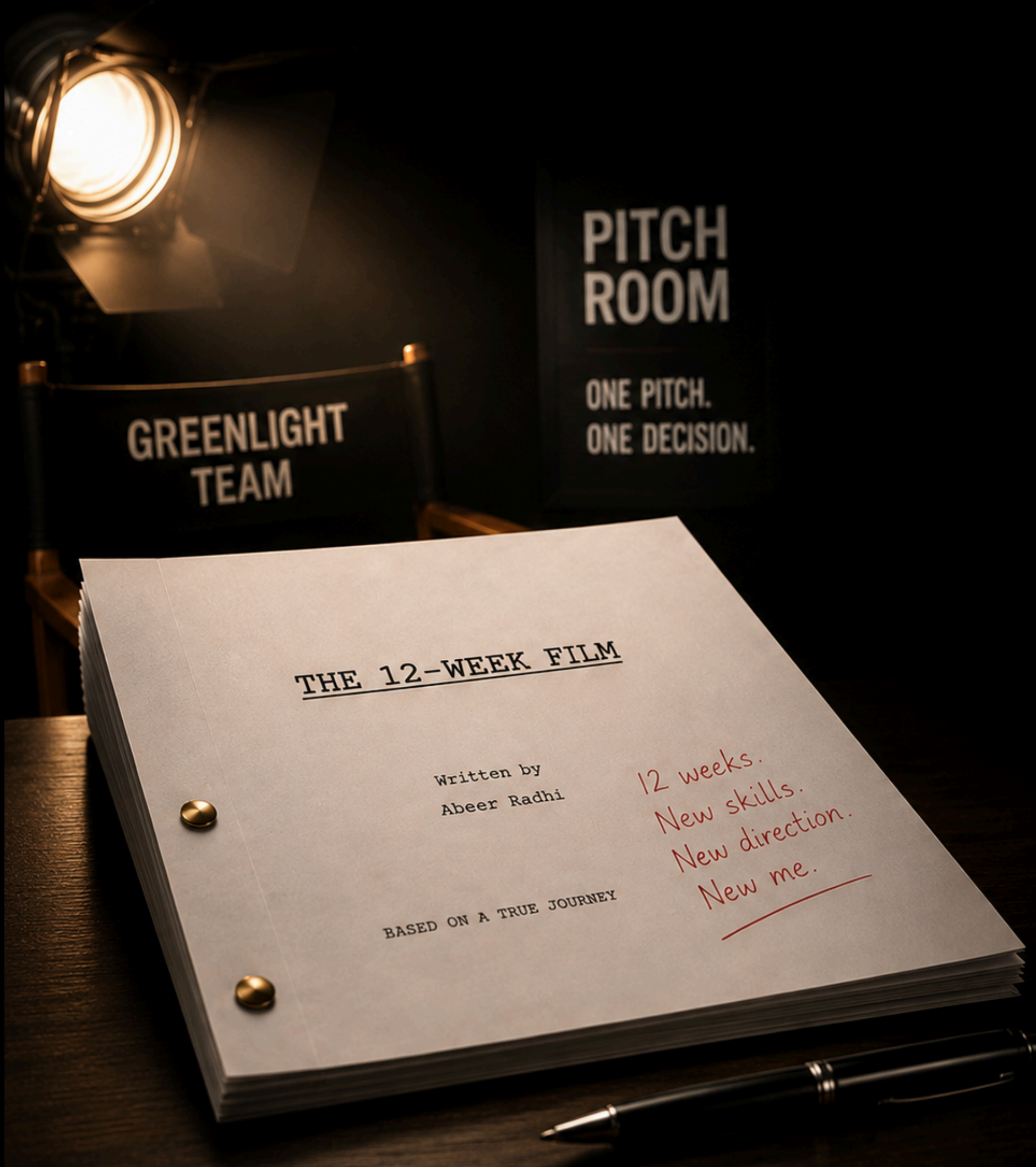
**What Relates to Reported
Revenue vs Budget?**

IMDb-TMDB Data Analysis | 2014–2023

Abeer Radhi

General Assembly - DAB 26

September 2026



THE 12-WEEK FILM

A story inspired by my journey through General Assembly

Imagine you're the greenlight team at a film production company. A writer walks in with a script based on her 12-week journey through General Assembly.



What genre is it?



What's the budget?



Who should play the main role?

WOULD YOU FUND THIS FILM?

Agenda

- 1 **Introduction**
- 2 Problem Statement
- 3 Objectives
- 4 Analysis and Findings

- 5 Recommendations
 - 6 Limitations
 - 7 Data Summary
-

Introduction

Film funding decisions are made before financial outcomes are known

This analysis explores which pre-release factors (genre, budget, cast experience, and creative team) are associated with a film's reported return.

Reported Return Measure:

$(\text{Reported Revenue} - \text{Production Budget}) \div \text{Production Budget}$

This measure excludes marketing, distribution, cinema/distributor shares, financing costs, and investor payments. It is not actual investor profit.

Target Audience

- **Primary:** Studio Greenlight Teams (who decide which film projects move forward)
 - **Secondary:** Investors, Financiers and Content Producers
-

Problem Statement

While film studios consider factors such as **genre, budget, and previous Lead experience** when assessing a **film's potential success** before funding it, these factors showed **weak relationships with reported return** across 2,465 selected IMDb-TMDB films released from **2014 to 2023**.

34.8% of 2,465 selected films,
below budget

\$11.05M average shortfall (858
films below budget)

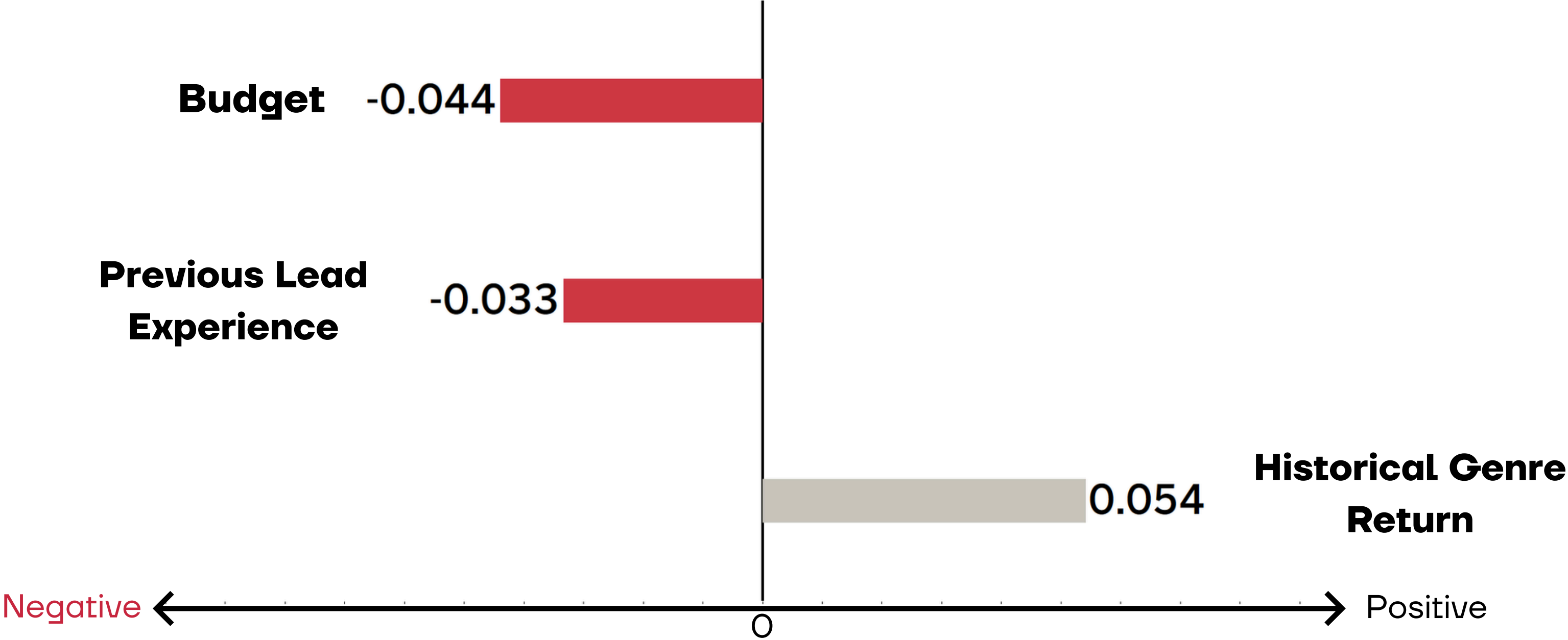
This creates uncertainty around **which factors** should receive greater consideration before **funding a film**.

Objectives

- 1 Determine the relationship between **budget, previous lead experience, genre**, and reported return
- 2 **Identify genres** with the lowest rates of films below budget
- 3 Determine how the rate of films below budget varies across **budget levels**
- 4 Compare reported return and below budget rates by **previous lead experience**
- 5 Compare reported return across **director and writer** prior history groups
- 6 **Combined Model:** Determine which factors show clearer relationships with reported return when considered together.

1

Pre Investment Factors Show Weak Return Relationships



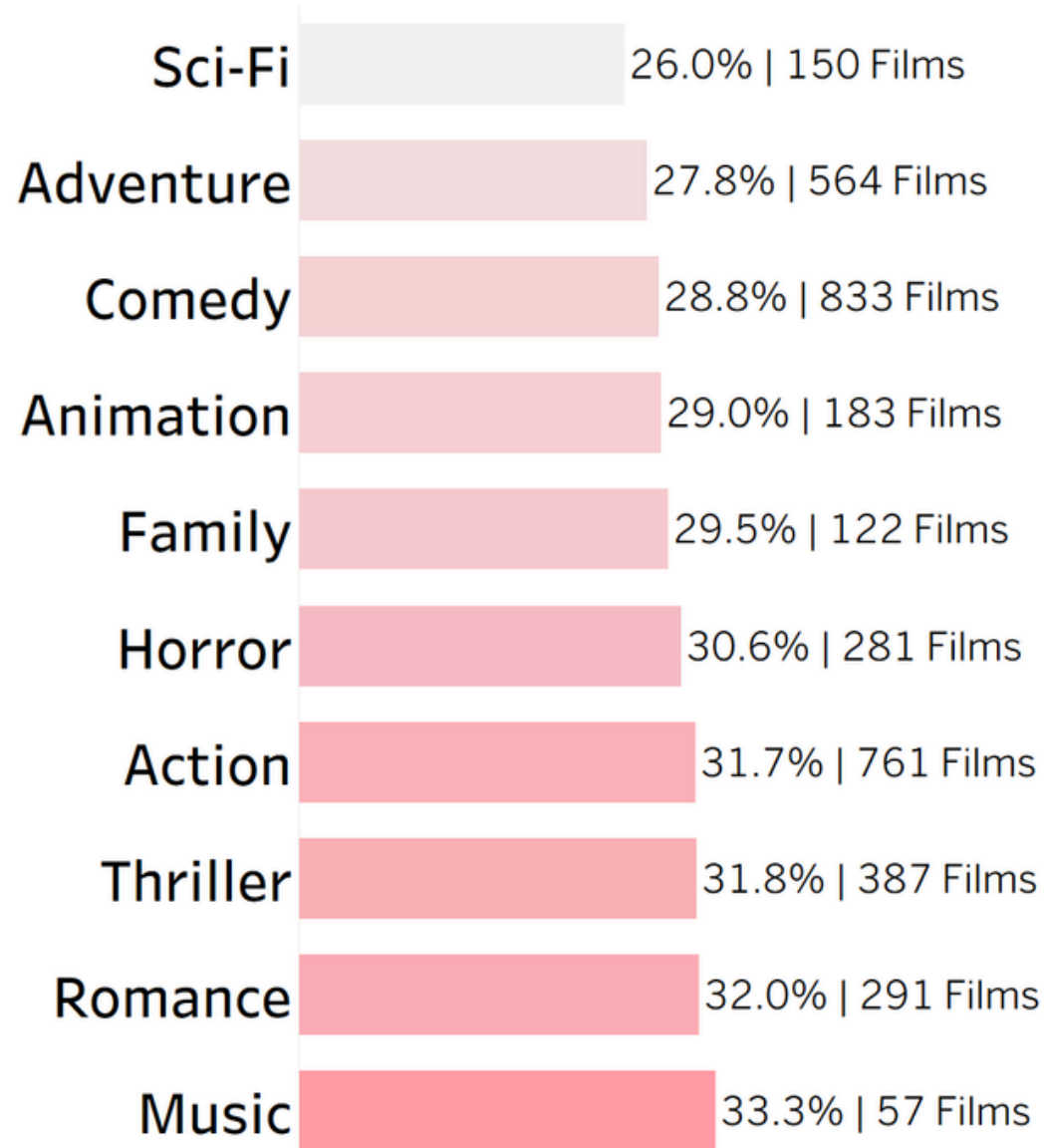
Correlation with Reported Return

Based on 2,465 films (2014–2023)

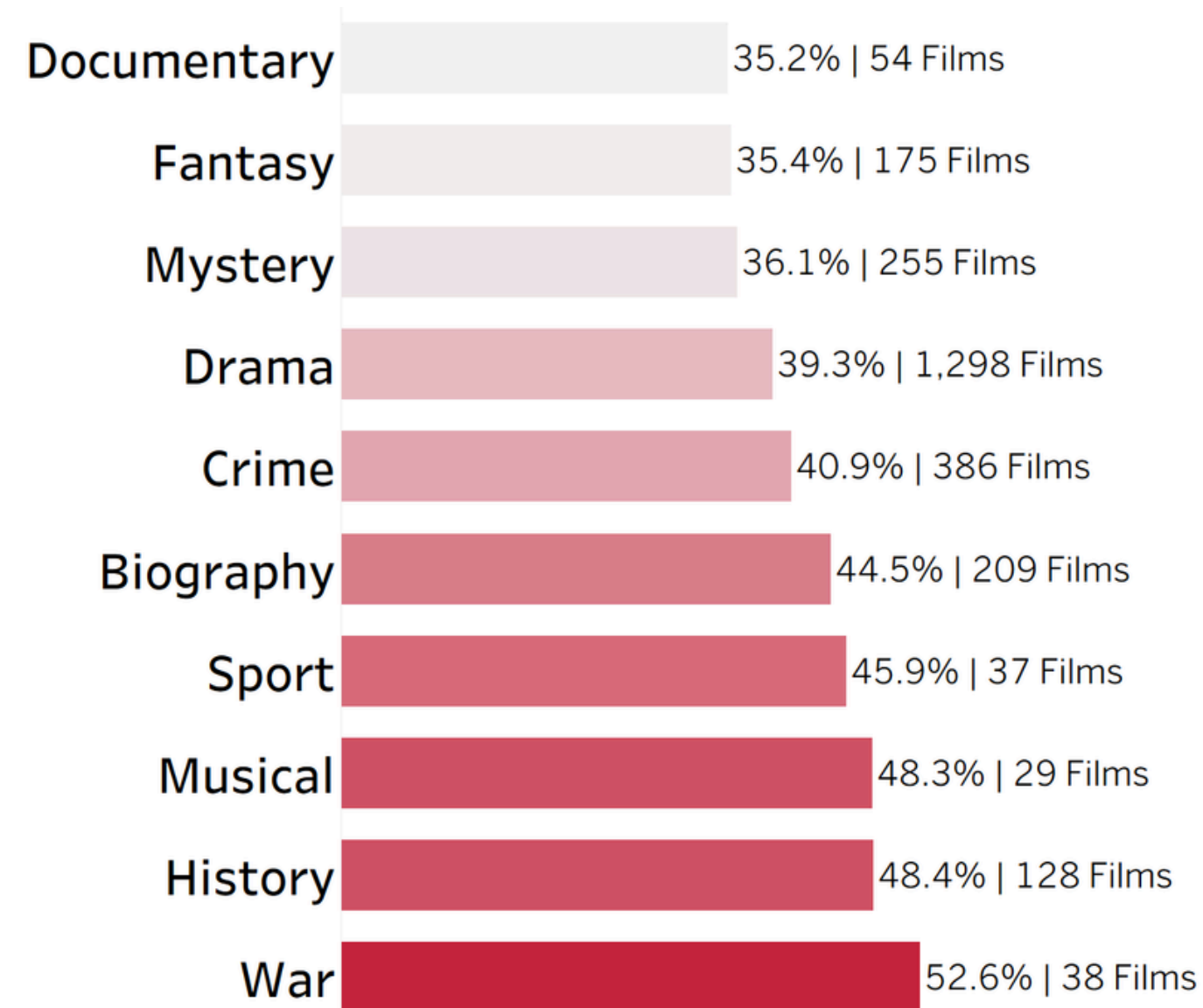
Values close to 0 mean a weak relationship. Sensitive to unusually large films.

Below Budget Rate by Genre

Lower Below Budget Rate



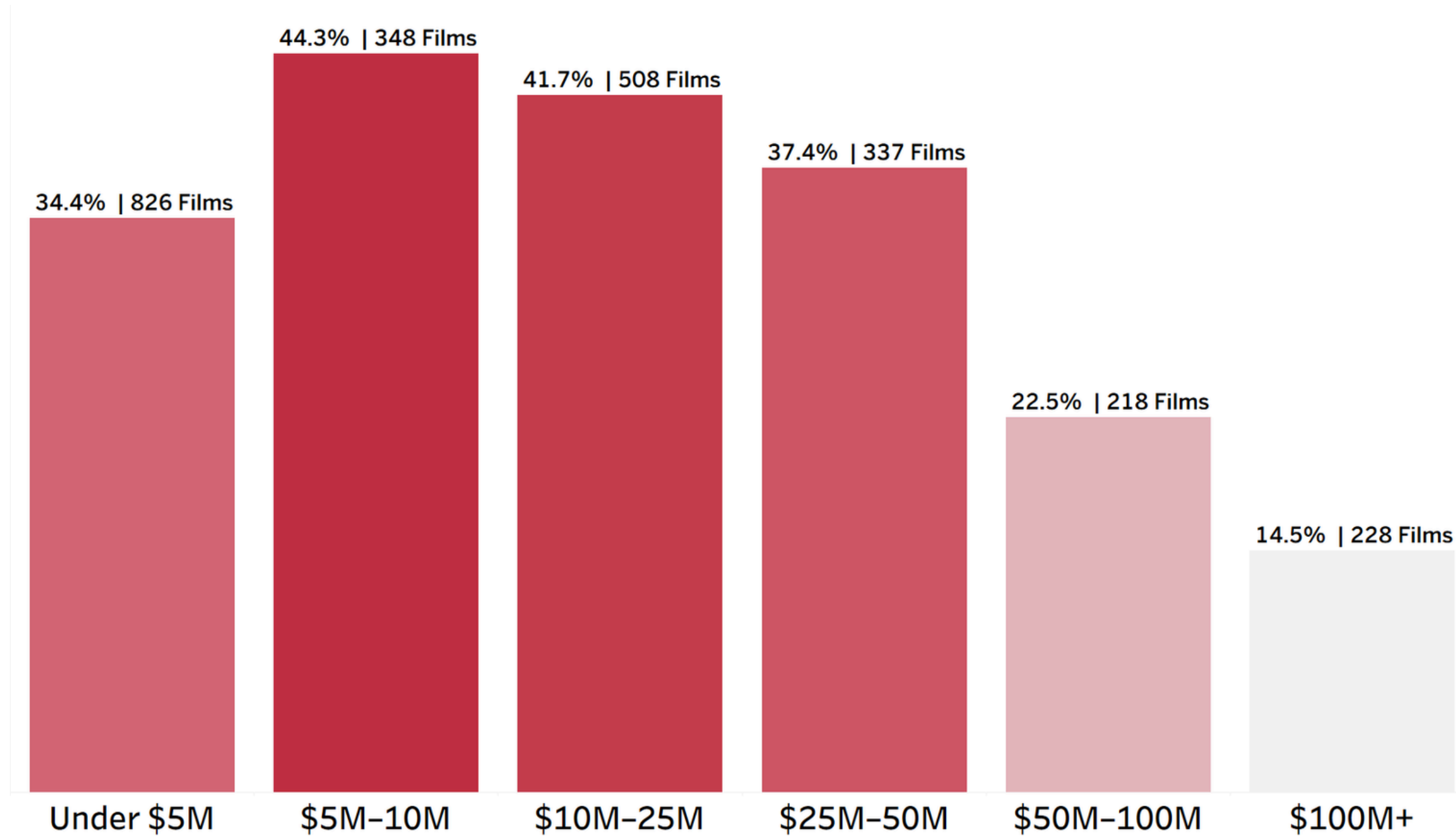
Higher Below Budget Rate



Genres shown have at least 25 films; films may belong to multiple genres.
Larger samples improve comparability but don't confirm reliability for any single genre.

3

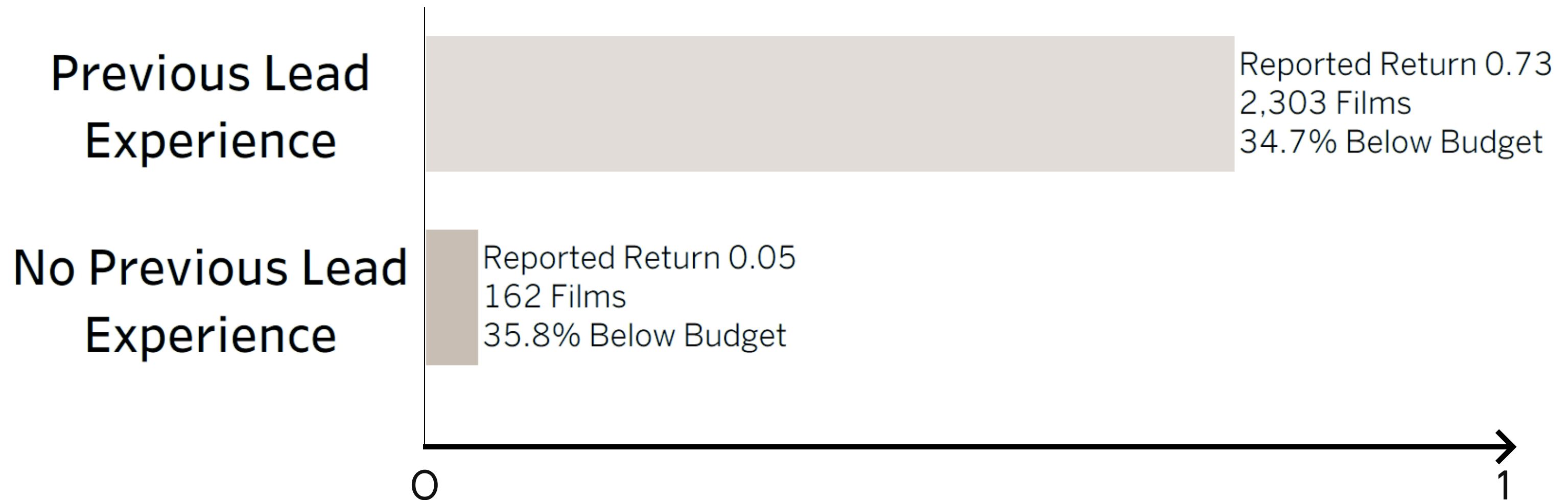
Larger Budgets Associate With a Lower Below Budget Rate



The percentage shows films below budget in each tier.
Association in this selected sample; not a funding rule.

4

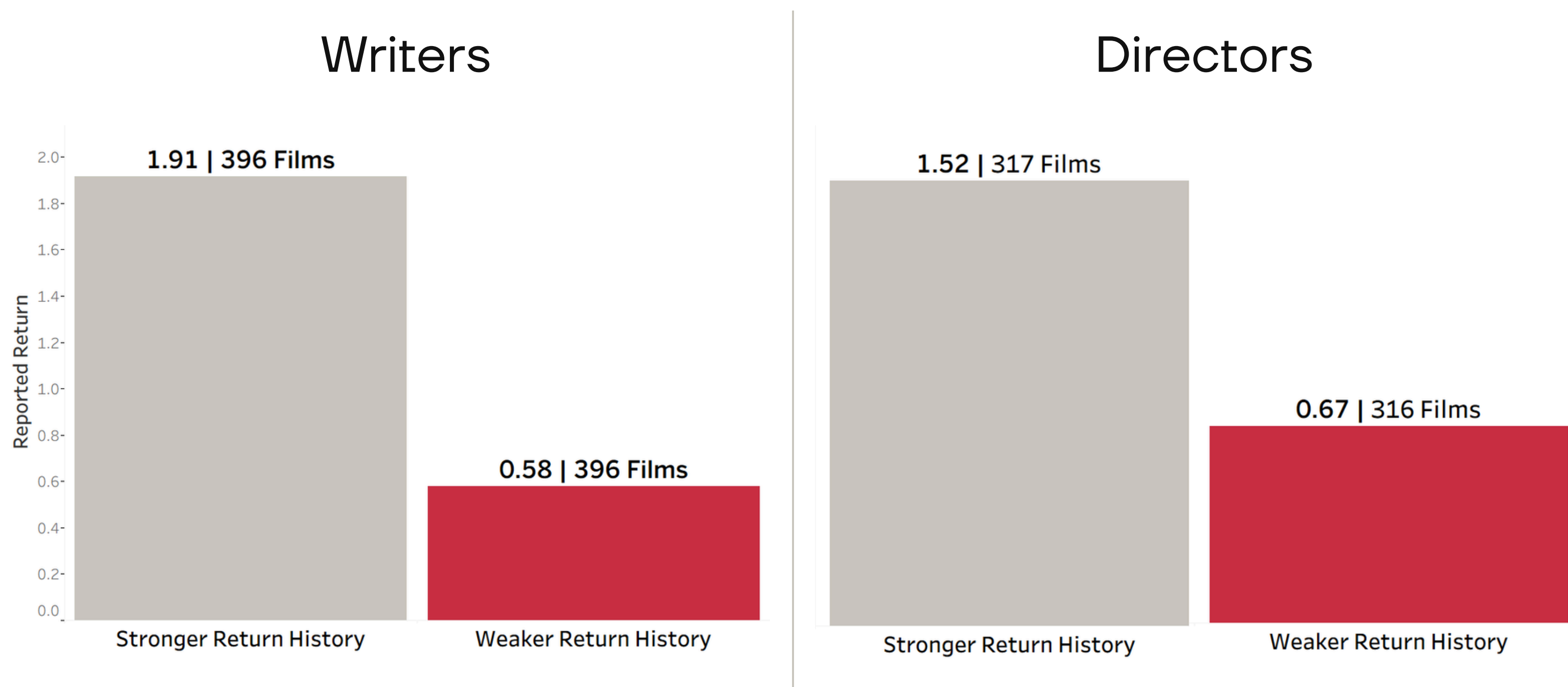
Lead Experience Shows Higher Return, Similar Below Budget Rate



Return differs substantially across the two groups.
Below budget rates remain nearly unchanged at 34.7% and 35.8%.

5

Stronger Creative Team History Shows Higher Reported Return



Prior history uses available earlier year films; same year releases excluded.

1.0 return = 100% above production budget.

Combined Factors Explain Limited Variation in Reported Return

Factors Considered Together with Production Company Control

Factor	Model Finding
Runtime	Clear relationship
Budget	Limited relationship
Previous Lead Experience	No clear standalone signal
Director Prior History	Limited relationship
Writer Prior History	Limited relationship
Genre Prior History	Limited relationship

Variation Explained: 9.1% | Adjusted: 7.5%

2,442 films | Reported return capped at 1st–99th percentiles | Not tested on new films

Recommendations

1. Combine Signals Before Greenlighting

Consider multiple factors rather than relying on one alone.

2. Compare Genre Funding Outcomes

Consider below budget rates when comparing genre options.

3. Treat Budget as Context

Compare budget levels without treating higher budgets as a funding rule.

4. Don't Use Lead Experience Alone

Previous lead experience showed no clear standalone signal.

5. Further Validate Creative Team History

Use broader career data before applying prior history to decisions.

Limitations

1. Simplified Return Measure

Reported return excludes major costs beyond production budget.

2. Selected Financial Sample

Only films with usable budget and revenue data were included.

3. Extreme Return Values

Unusually high returns may influence some results.

4. Limited Creative Team History

Director and writer history covers only part of the sample.

5. Model Not Externally Tested

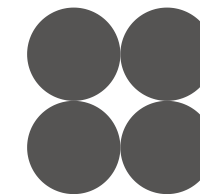
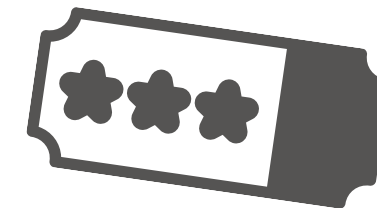
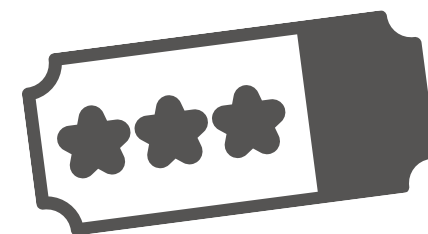
The model was not validated on unseen films.

Thank you!

Abeer Radhi

General Assembly – DAB 26

September 2026



Data Summary

DATA SOURCES

IMDb

Film details, genres, cast & crew information.

TMDB

Budget, revenue, production companies, and film information.

227,265 IMDb + 486,063 TMDB → 102,868 matched → 2,826 usable → 2,465 final films

FINAL ANALYSIS DATASET 2,465 films | 2014 to 2023 | 41 columns

IMDb + TMDB merged by title ID; filtered for usable reported budget and revenue.

KEY MEASURES

- **Reported Return** = $(\text{Revenue} - \text{Budget}) / \text{Budget}$ | values shown throughout this presentation represent the **median reported return** per group, not the average
 - **Below Budget Rate** = % of films with reported revenue below production budget
 - **Previous Lead Experience** = Whether the film's lead actor had prior lead experience
 - **Prior History** = Earlier available return history by genre, director, or writer
-